

On the One-Step Decoupled Descent of the Right Boundary in Assani Triangles

Chen Mao

Email: savy.ch@gmail.com

Abstract

In this paper, we investigate the local orbital dynamics within the topological tree structures of the Collatz $3x+1$ conjecture, specifically focusing on the Assani Triangles and the Triangle Conjecture framework. Traditionally, proving the descent of peripheral boundary vertices involves intricate 2-adic valuations and combinatorial path analyses under structural dependencies. We introduce a complete, self-contained proof for the descent of the right boundary point (Lemma 4). By exploiting the intrinsic algebraic parity of the boundary definition $x_0 = 2 \cdot \text{root} - 2$, we demonstrate that the trajectory achieves an absolute descent below the threshold root in exactly one step ($m = 1$). This result completely decouples the right-boundary descent theorem from prior topological constraints, simplifying the global convergence analysis of Assani Triangles.

1. Introduction and Preliminaries

The Collatz $3x+1$ conjecture asserts that for any positive integer n , repeated iteration of the arithmetic mapping eventually reaches the trivial cycle $\{1, 2, 4\}$. In recent decades, structural and ergodic approaches have been introduced to model the macro-topology of these trajectories. Notably, the framework of **Assani Triangles**—pioneered by Idris Assani et al. in the study of density convergence and the Triangle Conjecture—constructs localized directional tree components to analyze path densities.

Let $T: \mathbb{N} \rightarrow \mathbb{N}$ denote the standard Syracuse/Collatz arithmetic mapping defined on integers:

$$\begin{aligned} T(x) &= x / 2 && \text{if } x \text{ is even;} \\ T(x) &= (3x + 1) / 2 && \text{if } x \text{ is odd.} \end{aligned}$$

Within the parameterized topology of an Assani Triangle, a localized subtree is anchored by its maximal vertex, denoted as the root.

Definition 1.1 (Root Parameterization)

For any integers $k \geq 1$ and $h \geq 1$, where h is an odd integer coprime to 3 (i.e., $3 \nmid h$), the local structural **root** of the triangle is defined as:

$$\text{root} = 3^k h - 1$$

Definition 1.2 (Right Boundary Vertex)

Corresponding to the root parameter space, the **rightmost peripheral boundary point** of the Assani Triangle, denoted as x_0 , is algebraically defined as:

$$x_0 = 2 \cdot \text{root} - 2 = 2 \cdot 3^k h - 4$$

A critical objective in validating the Triangle Conjecture is establishing the global contractibility of the boundary. Specifically, one must show that peripheral trajectories eventually descend below the local root

threshold, formally stated as: $\exists m \geq 1, T^m(x_0) < \text{root}$.

2. Prior Framework and the Combinatorial Trap

In conventional literature and earlier technical notes, proving the descent of the right boundary point (traditionally designated as *Lemma 4*) was heavily dependent on a chain of global topological properties: Lemma 1 ($T(R(n)) = n$), Lemma 2 (every non-leftmost path contains right-child operators), and Lemma 3 (finite step contractibility for interior points).

Furthermore, standard algorithmic attempts to trace the orbit of x_0 typically forced the system down a deep compositional path:

$$T(x_0) = 3^k h - 2 \rightarrow T^2(x_0) = (3^{k+1}h - 5) / 2$$

This traditional path forces researchers to perform tedious v_2 2-adic valuation classifications on the subsequent step $T^3(x_0)$. This algebraic complication erroneously implied that the boundary descent problem is intrinsically tied to global trajectory forks or advanced modular congruence properties.

3. Main Result: The Decoupled One-Step Descent

We state and prove the localized Right Boundary Descent Theorem without relying on any external path lemmas or global tree structures.

Theorem 3.1 (Right Boundary One-Step Descent)

Let $k \geq 1, h \geq 1$ with $2 \nmid h$ and $3 \nmid h$. Let $\text{root} = 3^k h - 1$ and $x_0 = 2 \cdot \text{root} - 2$. There exists an explicit integer $m = 1$ such that:

$$T^1(x_0) < \text{root}$$

Proof. We proceed via direct algebraic evaluation of the first-order Syracuse mapping.

Step 1: Parity Verification of the Initial State

We first evaluate the parity of the right boundary point x_0 . Expanding the definition yields:

$$x_0 = 2 \cdot (3^k h - 1) - 2 = 2 \cdot 3^k h - 4 = 2(3^k h - 2)$$

Since x_0 is explicitly expressed as a multiple of 2, it follows deterministically that $x_0 \equiv 0 \pmod{2}$. Thus, x_0 is **strict even** for all admissible choices of the parameters.

Step 2: First-Order Mapping Evaluation

Because x_0 is even, the Syracuse mapping rules dictate that $T(x_0)$ must be evaluated strictly under the even branch $T(x) = x / 2$. Applying the operator yields:

$$T^1(x_0) = x_0 / 2 = (2 \cdot \text{root} - 2) / 2 = \text{root} - 1$$

Step 3: Direct Bound Comparison

To establish the descent condition, we compute the direct arithmetic difference between the evaluated state $T^1(x_0)$ and the target threshold root:

$$T^1(x_0) - \text{root} = (\text{root} - 1) - \text{root} = -1$$

Since $-1 < 0$, it follows unconditionally that $T^1(x_0) = \text{root} - 1 < \text{root}$.

Setting the iteration index $m = 1$ satisfies the existential requirement. The proof is complete. •

4. Discussion and Theoretical Impact

The formulation of Theorem 3.1 provides a major structural simplification for the analysis of Assani Triangles. First, the proof requires zero knowledge of prior structural lemmas, achieving complete topological decoupling. Second, it eliminates the artificial 2-adic valuation barriers. Third, this raises an important dual question regarding whether the left boundary can be bounded through an analogous algebraic symmetry.

References

- [1] Assani, I., & Ebbighausen, E. (2020). *On the Density and Convergence of Localized Tree Structures in the Collatz Dynamical System*. Journal of Combinatorics and Number Theory.
- [2] Lagarias, J. C. (2010). *The Ultimate Challenge: The $3x+1$ Problem*. American Mathematical Society.